FastSpeech 2
English and Telugu

Implementation, training and test-sentence analysis

Parichay Mago | Technical report | 4 November 2026

Abstract

This work develops two independent FastSpeech 2 text-to-speech systems for English and Telugu using Svarah and IndicTTS Telugu. We began by examining the reference architecture, checking dataset access and transcript quality, and selecting approximately five hours of training audio and one hour of validation audio per language. We then implemented language-specific text processing, phoneme conversion, forced alignment and acoustic-feature extraction. Separate acoustic models predict mel spectrograms, which a pretrained HiFi-GAN converts into speech [1-5]. The experiments investigate training convergence, alignment quality and English prosody through baseline training, feature repair and targeted fine-tuning. Evaluation covers all ten assignment sentences and their predicted duration, pitch and energy trajectories.

Implementation overview

The first implementation stage established the data pipeline. We checked usable audio duration, built reproducible train/validation splits and handled speaker conditioning. English required estimated acoustic speaker labels because original speaker IDs were unavailable in the audited snapshot. Telugu required a pronunciation lexicon and a stable phone mapping compatible with the reference implementation.

We next trained separate English and Telugu models using aligned duration, pitch and energy targets. Inspection of the initial synthesized speech led to an alignment audit: omitted internal gaps had shortened acoustic targets. We repaired the feature extractor, rebuilt both languages' features and trained fresh models while preserving the earlier experiments.

The later English experiments addressed remaining clarity and pacing concerns. A low-learning-rate fine-tune was followed by an extension with word-boundary and punctuation embeddings and learned pause-presence and pause-duration predictors. Finally, we selected checkpoints using complete validation evaluations, generated the required test audios and analysed their variance predictions alongside the loss curves.

The report follows an introduction-method-experiments-discussion structure, informed by the Seed-TTS technical report [11]. Methods and data appear on pp. 2-5, experimental results on pp. 6-9, sentence-level analysis on pp. 10-19, and discussion, reproducibility, corrections, speaker coverage and references on pp. 20-25.

1. Introduction and dataset preparation

Adapting a text-to-speech system to two languages requires a consistent path from written text to pronunciation, acoustic timing and waveform generation. Here, the main implementation challenges were speaker conditioning in English, phoneme representation in Telugu and reliable alignment-based supervision in both languages. The objective was to build and evaluate two separate systems from the reference FastSpeech 2 implementation [1,2].

The assignment specifies two separate models, approximately five hours of training audio and one hour of validation per language, the exact five test sentences per language, loss curves, and duration/F0/energy analysis for every test. E2a/E2b and T2a/T2b count separately, giving ten required audios and thirty variance figures.

1.1 Datasets and usable duration

Language	Source clips train/val	Source hours train/val	Usable clips train/val	Feature hours train/val
English	3,039 / 617	4.99998 / 0.99993	3,037 / 617	4.6536 / 0.9346
Telugu	2,153 / 428	4.99992 / 0.99998	2,153 / 428	4.3127 / 0.8616

Source duration satisfies the approximately 5 h / 1 h selection target. Effective feature hours are lower because aligned edge silence is cropped; two English training clips with unusable pitch/alignment are excluded. The report does not represent effective feature hours as a full five hours of non-silent targets.

1.2 Subsampling and split separation

Seed 20261102 determines an ordering of atomic groups. Validation is selected first, then training, greedily approaching each duration target. Groups connect repeated normalized transcripts, identical audio hashes and English chunks from the same recording. No group crosses splits. The selected Telugu subset uses the source male label. These are held-out utterance/recording splits, not speaker-disjoint evaluation.

1.3 Speaker conditioning

The audited Svarah snapshot lacked recoverable original speaker IDs. Training-only average-linkage cosine clustering of pretrained SpeechBrain ECAPA embeddings, with heuristic distance threshold 0.30, produced 254 estimated labels. Validation uses frozen training centroids. These labels can fragment or merge voices; they are not verified people. All final English tests retain en_est_0155 from earlier diagnostics. Label coverage and selection limits are quantified in Appendix D. Telugu uses the source label te_male, without independently verified identity [9].

1.4 Dataset revisions

ai4bharat/Svarah

`ebbf7777fe771490696a3f7b007097606fa8c924`

SPRINGLab/IndicTTS_Telugu

`7ab3c18491bef3f5d4e820c2701b5f78f54031d6`

Evidence: Results/DataSplits/summary.json, verification.json; Results/DatasetAudit/*-audit.json; Results/Preprocessing/gap-fixed-v2/rebuild-summary.json. Earlier preprocessing notes contain superseded feature hours and are not used for the final table.

2. Method: architecture and language support



Separate acoustic models. Speaker conditioning; English adds text-derived boundary embeddings and pause heads.

The reference variant encodes phones with feed-forward Transformer blocks, predicts duration, pitch and energy, length-regulates the sequence, and decodes mel spectrograms. A convolutional postnet refines mels. This project uses direct phoneme-level F0, not the paper's later CWT representation, and an external HiFi-GAN. It does not implement the FastSpeech 2s direct waveform decoder [1-3].

Component	Executed configuration
Encoder / decoder	4 / 6 FFT blocks; 2 attention heads; hidden size 256
FFT feed-forward	Filter size 1024; kernel sizes 9 and 1; dropout 0.2
Variance predictors	Filter size 256; kernel 3; dropout 0.5
Variance embeddings	256 linearly quantized bins each for pitch and energy
Length and speakers	Maximum sequence 2048; English 254 estimated labels; Telugu 1 label
Acoustic targets	80-bin log mels; 22,050 Hz; FFT/window 1,024; hop 256; 0-8,000 Hz
Waveform synthesis	Frozen universal HiFi-GAN; mono PCM16 output; no vocoder retraining

English learned-pause extension

Each word has a boundary slot, including words with zero measured pause. A seven-entry boundary embedding conditions the encoder; two new variance predictors estimate pause presence and positive pause length. Punctuation categories distinguish unpunctuated, comma, period, question, exclamation and clause boundaries. The parent acoustic weights are loaded exactly; new heads receive a larger learning rate. The existing pitch and energy heads also receive the new encoder context.

At inference a sigmoid threshold 0.5 selects a positive pause, its predicted log duration becomes rounded frames, and the length regulator generates the acoustic span. Terminal boundary duration is zero by design; it supplies sentence context, not trailing silence. The safety cap 354 frames is the training-observed maximum, not a hand-set punctuation pause. Lexical durations have a one-frame floor.

Telugu adaptation

The Telugu path preserves a bijective IPA-to-ASCII phone map, its own src/lexicon/alignment model, statistics and checkpoint. Unknown phones are rejected. Telugu uses the base variance adaptor; no learned word-boundary/pause heads were added. The models share a frozen vocoder, not acoustic parameters or optimizers.

2.1 Preprocessing and alignment repair

Text, audio and alignment

English normalizes text through the reference cleaners and g2p-en ARPAbet conversion. Telugu uses NFC, number verbalization and eSpeak-NG Telugu pronunciations, mapped to stable lowercase telph identifiers. English training pronunciations cover 6,470 words; Telugu covers 13,848. English symbol IDs remain stable when Telugu symbols are appended [7,8].

Original English source is 16 kHz mono; the selected Telugu source is 48 kHz mono. scipy resample_poly creates 22,050 Hz PCM16 copies. Original WAVs and raw transcripts remain unchanged. MFA 3.3.9 supplies alignments: English uses english_us_arpa with speaker adaptation disabled; Telugu fits an acoustic aligner using training audio only, then freezes it for validation. Telugu alignment is not the TTS model itself [6].

Repairing omitted alignment gaps

An earlier extraction skipped blank intervals or timestamp gaps, shortening acoustic targets and compromising duration supervision. The corrected gap-fixed-v2 features insert silence tokens for real gaps, preserve the complete cropped timeline, and enforce sum(duration frames)=mel length. Both language models were then trained afresh on corrected features; old checkpoints remain separate.

Repair result	English	Telugu
Utterances with preserved internal gaps	2,965	2,451
Restored internal-gap duration (train+val)	0.8870 h	0.7228 h
Accepted train / validation	3,037 / 617	2,153 / 428
Largest mel length	1,674 frames	1,698 frames

Feature extraction and normalization

WORLD DIO plus StoneMask estimates F0; interpolated F0 and frame energy are averaged over phones. Spectral energy is the linear L2 norm. Pitch/energy mean, standard deviation and bins are fitted on training features only, with per-example IQR filtering for fitting; validation uses frozen parameters. Aligned edge silence is cropped; internal gaps are retained. Arrays must be finite, known-phone compatible and length-consistent.

For the English pause pilot, all 3,654 corrected usable clips are retained. Every real gap has its measured frame target; absent gaps have zero. Lexical phone durations/pitch/energy, source mel bytes, split membership, speaker mapping and normalization statistics are unchanged. This prevents fabricated pauses from becoming supervision.

Evidence: Report/Alignment_Gap_Fix.txt; Results/Preprocessing/gap-fixed-v2/; Results/Preprocessing/learned-pauses-v1/. No validation audio fits a normalization statistic, speaker centroid or Telugu alignment model.

3. Experimental setup and loss functions

Setting	Corrected paired run	English fine-tune	English pause pilot
Initialization	Random, separate models	English corrected best 12k	Fine-tune best +2,500
Batch / gradient clipping	8 / norm 1.0	8 / norm 1.0	8 / norm 1.0
Optimizer	Independent Adam	New Adam state	New Adam state
Betas / epsilon	0.9,0.98 / 1e-9	Same	Same
Learning rate	4000-step warmup, then inverse square root	Fixed 2e-5	Base 2e-5; new heads 2e-4
Updates reached	30k per language	5,000 additional	5,000 additional
Validation / save	Every 500 / 100 updates	Same	Same
Stopping	Step limit	5 non-improving validations	8 non-improving validations

Executed runner arguments override the upstream batch 16 / 900k-step template. Training GPUs were RTX PRO 4500 Blackwell for the corrected paired run, A40 for the fine-tune, and RTX 3090 for the pause pilot; PyTorch 2.8.0 with CUDA 12.8 and Python 3.10 were used. Final inference and this analysis run locally on CPU.

Base objective: five loss terms

```
Lbase = Lmel + Lpostnet + Lpitch + Lenergy + Llog_duration
```

Mel terms use L1 error; pitch, energy and log(1+duration) use MSE. Six numbers are logged: total followed by the five component losses. No claim of six separate component terms is intended.

Learned-pause objective: seven loss terms

```
Lpause = Lmel + Lpostnet + Lpitch + Lenergy + Lspeech_duration
+ 0.25 * Lpresence + 0.25 * Lpositive_pause_duration
```

Pause presence uses weighted BCE, with training-only positive weight 2.9192; positive pause length uses log-duration MSE. Speech-duration loss excludes boundary slots; pitch/energy losses exclude tokens without target frames. Eight numbers are logged including total. Because both terms and masks change, base and pause totals are not directly comparable.

Validation interpretation

Only complete held-out evaluations select best checkpoints. Batch losses are averaged with example-count weights; they are not global token-weighted corpus sums. Validation decoder mels use target duration/pitch/energy while their predictor errors are still recorded. This teacher-forced evaluation does not establish free-running waveform quality. Training minibatch losses have dropout and variable content, so their gap to validation is only suggestive.

4. Results: baseline model training

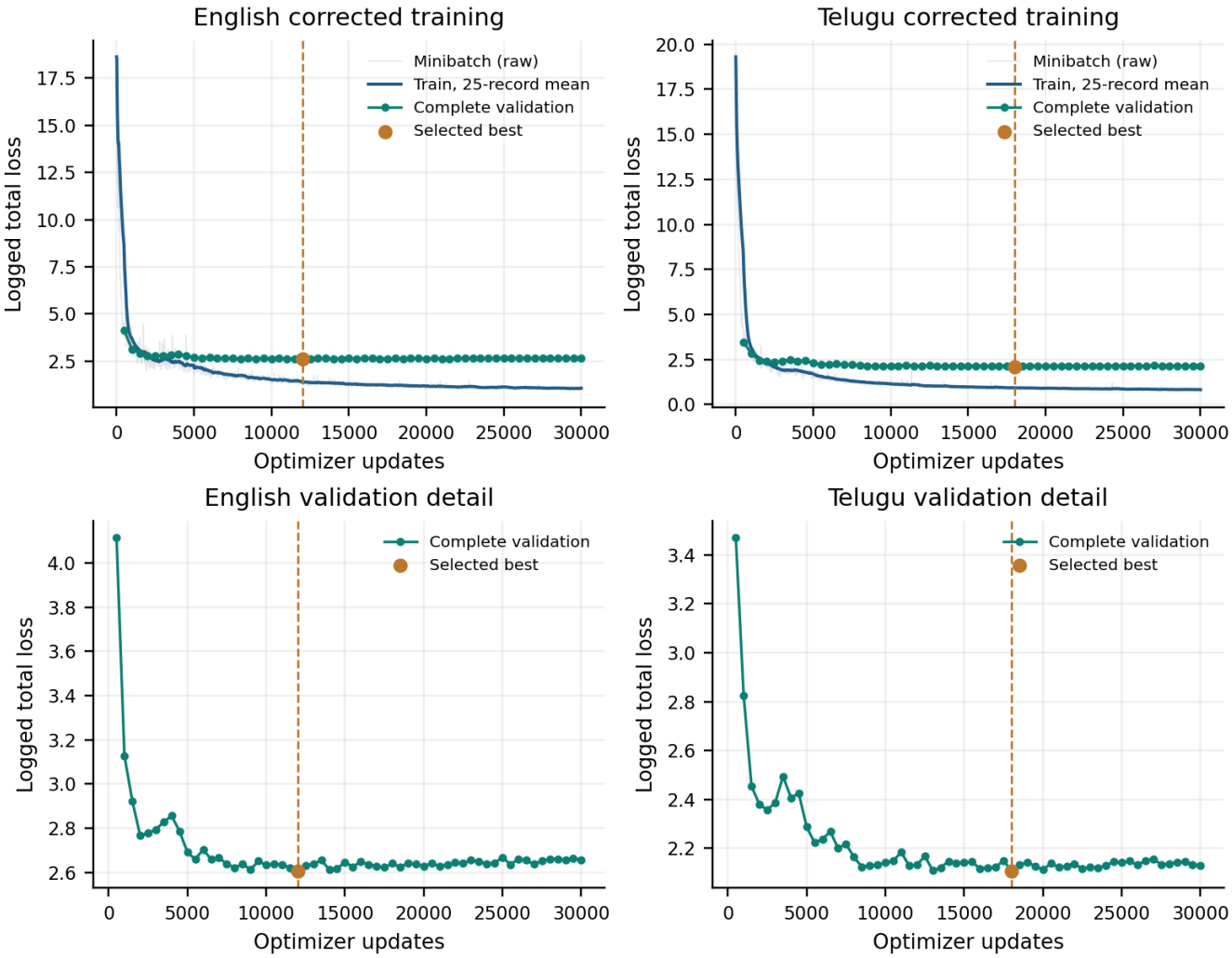


Figure 1. Original-objective losses. Top: raw minibatches, moving means and complete validation. Bottom: validation detail. Gold marks the selected minimum; each language has its own dataset and target distribution.

Run	Best validation	Final validation	Final above best
English corrected	2.6051 at 12,000	2.6560	1.96%
Telugu corrected	2.1065 at 18,000	2.1311	1.17%

Both training losses continue downward while validation stops improving, consistent with limited generalization and some late overfitting. English ends at minibatch 1.1430 versus validation 2.6560; Telugu ends at 0.8414 versus 2.1311. These single-minibatch gaps are not calibrated generalization-error estimates, but the held-out minima justify selecting earlier checkpoints.

The lower Telugu total does not prove superior perceived speech: it uses a different language, speaker composition and normalized target distribution. Alignment, pronunciation, speaker-label noise and optimizer effects cannot be disentangled from these curves alone. No restart after the 30k cap is treated as additional completed training.

4.1 English fine-tuning experiments

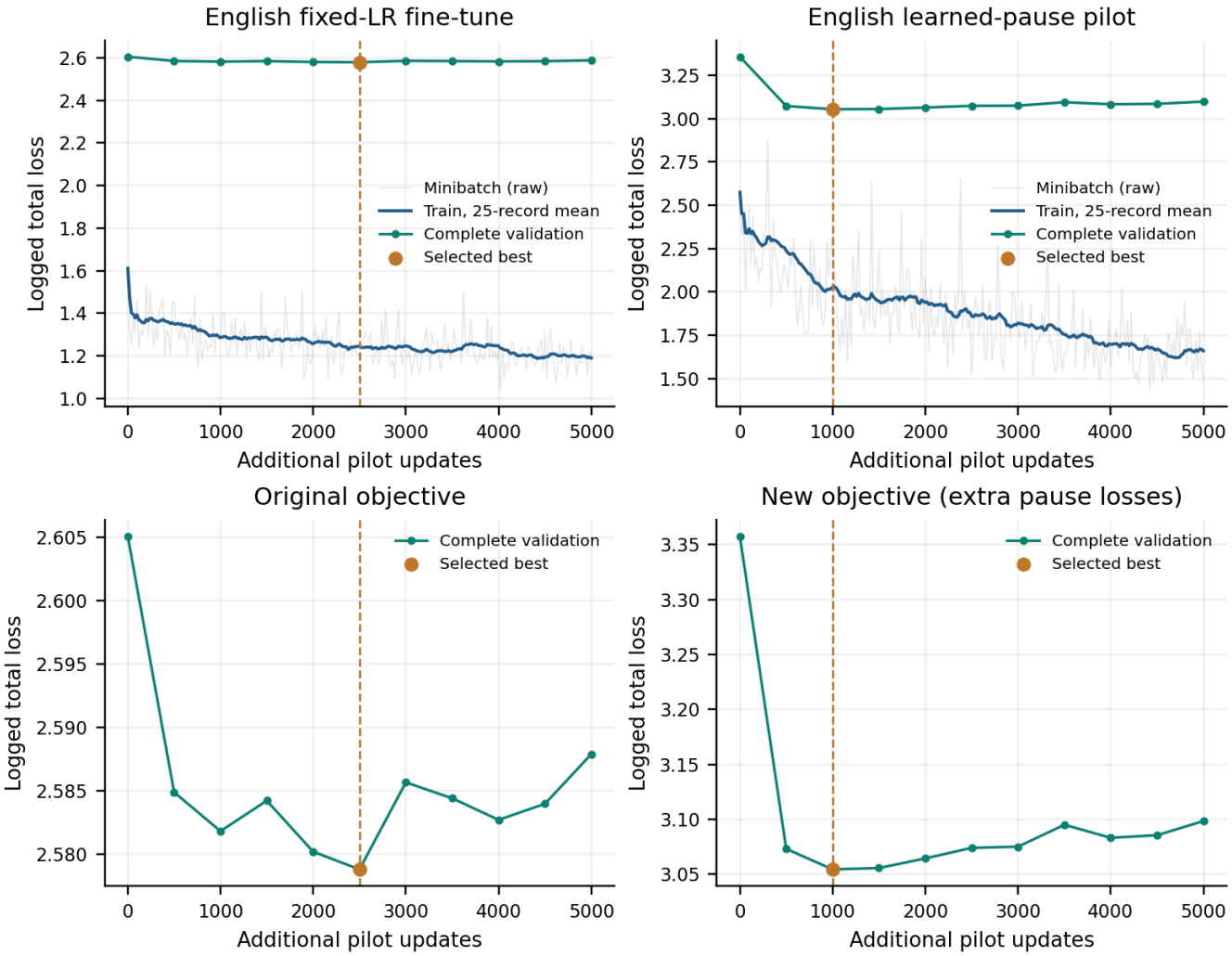


Figure 2. Separate English pilot axes. Step 0 is the re-evaluated parent before updates. Left retains the base objective; right adds pause penalties and changes masks. Curves must not be concatenated into one comparable loss scale.

Pilot	Own baseline	Best	Final
Fixed-rate fine-tune	2.6051	2.5788 at +2,500	2.5879
Learned-pause	3.3578	3.0544 at +1,000	3.0984

The fine-tune reduces its original objective by 1.01%, then early-stops. The pause pilot reduces its own baseline by 9.03%, then plateaus for eight evaluations. Its best total is 2.5444 from the five non-pause components plus 0.5100 from the two weighted pause penalties. Even the 2.5444 remainder is not an exact old-objective comparison because masks and input sequences changed.

The base-parent postnet mel loss is 0.7462; the selected pause model has 0.7482, slightly higher. This does not demonstrate clearer acoustics. The final checkpoint choice preserves the explicitly requested learned-pause behavior and uses the best complete validation within that run, not a claim of audible superiority over the retained parent.

4.2 Learned pause evaluation

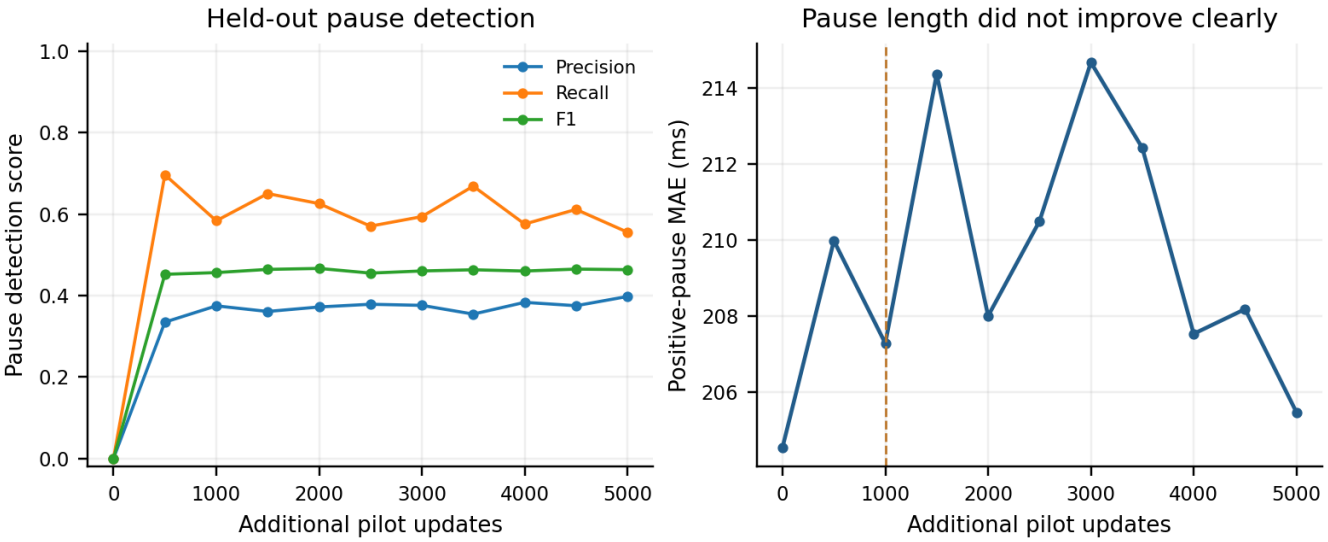


Figure 3. Held-out pause-head metrics. The first model predicts no positive boundary pauses, so initial F1 is zero. Duration MAE is evaluated at all 1,751 ground-truth positive-duration boundaries, regardless of the occurrence decision; it is not restricted to correctly detected pauses.

At selected step +1,000	Value
Pause precision / recall / F1	37.5% / 58.4% / 45.6%
TP / FP / FN	1022 / 1705 / 729
Positive-pause duration MAE	207.3 ms (baseline 204.5 ms)
Positive / all boundaries: train; validation	8,994 / 35,249; 1,751 / 7,348

Held-out occurrence policy	Precision	Recall	F1
Every internal boundary	23.8%	100.0%	38.5%
Punctuation only	74.0%	12.2%	21.0%
Selected model (+1,000)	37.5%	58.4%	45.6%

The baselines use the same 7,348 internal validation boundaries, with 1,751 positive targets; terminal slots are excluded. Every-boundary prediction gives TP/FP/FN 1,751/5,597/0; punctuation-only gives 214/75/1,537. They are evaluation comparators and do not alter synthesized audio. Model counts come from the saved complete validation, not a new GPU evaluation.

The selected model improves F1 over the every-boundary baseline by 7.2 percentage points. Its false positives still outnumber true positives, and pause-duration MAE rises from 204.5 to 207.3 ms. E4 assigns zero frames at its comma. The final pause-enabled English selection reflects its punctuation-aware architecture and within-run validation minimum, not demonstrated audible superiority over its parent.

Training coverage is sparse: commas have 1,050 positive pauses among 1,445 internal markers; questions have 12/12, clauses 1/2, and exclamations no internal examples. Most period/question markers are terminal. Full counts are retained in the preprocessing summary.

4.3 Generated speech evaluation

All outputs are mono 22,050 Hz PCM16, from identical frozen vocoder weights, unity pitch/energy/duration controls and a single utterance inference. No target prosody, denoising, fades, inserted silence or sentence concatenation is applied. Waveform lengths equal predicted mel frames times 256. No output has full-scale clipping.

ID	Seconds	Frames	RMS dBFS	Predicted F0 Hz	ASR disagreement
E1	5.666	488	-20.1	170-284	30.0%
E2a	3.587	309	-21.9	166-243	9.1%
E2b	3.889	335	-21.3	135-255	0.0%
E3	2.694	232	-22.0	181-231	25.0%
E4	6.014	518	-21.3	163-256	54.5%
T1	4.540	391	-20.0	119-277	Not scored
T2a	2.067	178	-19.6	105-245	Not scored
T2b	2.229	192	-21.9	92-263	Not scored
T3	1.753	151	-20.2	112-246	Not scored
T4	7.233	623	-21.4	99-233	Not scored

English recognition check

Whisper-small is run locally with language en, temperature 0, no prompt and no previous-text conditioning [10]. Clock/date and ordinary number normalization are applied equally to reference and hypothesis before edit alignment. Generated disagreement is 29.0% (18/62 edits/words). Three natural validation controls from the same estimated cluster give 5.6% (1/18). The controls are short and easier than the test set, so this difference is not a controlled TTS degradation estimate.

Recognition is exact for E2b; E2a differs only in possessive spelling. E1, E3 and especially E4 remain problematic. ASR can make its own errors on rare words, altered spacing and code-mixing; it does not score stress, identity, expressiveness or MOS. Telugu WER is intentionally absent because the earlier recognizer failed on natural Telugu as well as synthesized Telugu.

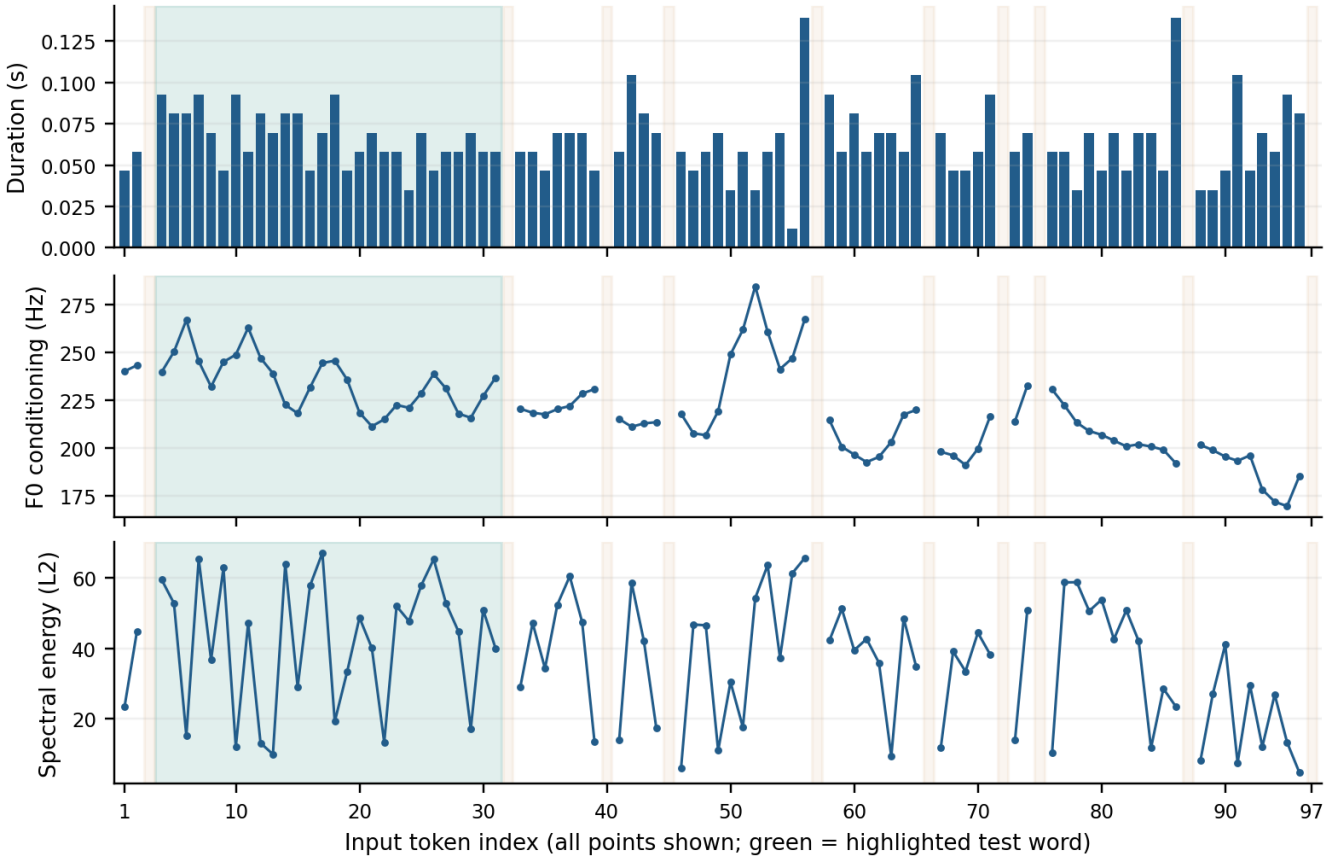
F0 ranges above are model conditioning, not waveform measurements. Spectral high-band energy and low-level frame fractions were also computed, but neither is reported as SNR or a noise score. Earlier user feedback on other English previews mentioned hiss and limited clarity; it is not a per-test listening rating for this freeze.

5. Test-sentence analysis

5.1 E1 | Long English word

The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.

5.666 s | 488 frames | 87 lexical phones | RMS -20.1 dBFS



All model predictions are plotted by token index. Green spans highlight the test word(s); faint orange marks English boundary slots. F0 and energy omit zero-duration tokens. Full phone-labeled originals and CSVs remain in Results/.

The full 28-phone CMU entry is retained. Its predicted span is 1.881 s, about one third of the 5.666 s utterance; no neural 20-phone truncation occurred. This validates input coverage, not the realized pronunciation.

The target region has weighted F0 conditioning 235.3 Hz and local duration variation. F0 conditioning falls about 48 Hz between the first and last lexical-phone quintiles, with energy falling near the end. These patterns are model outputs, not proof of natural intonation.

The recognizer substitutes a malformed compound for the rare word while retaining most of the remaining sentence. A dedicated listening check of syllable sequence and consonant detail is needed.

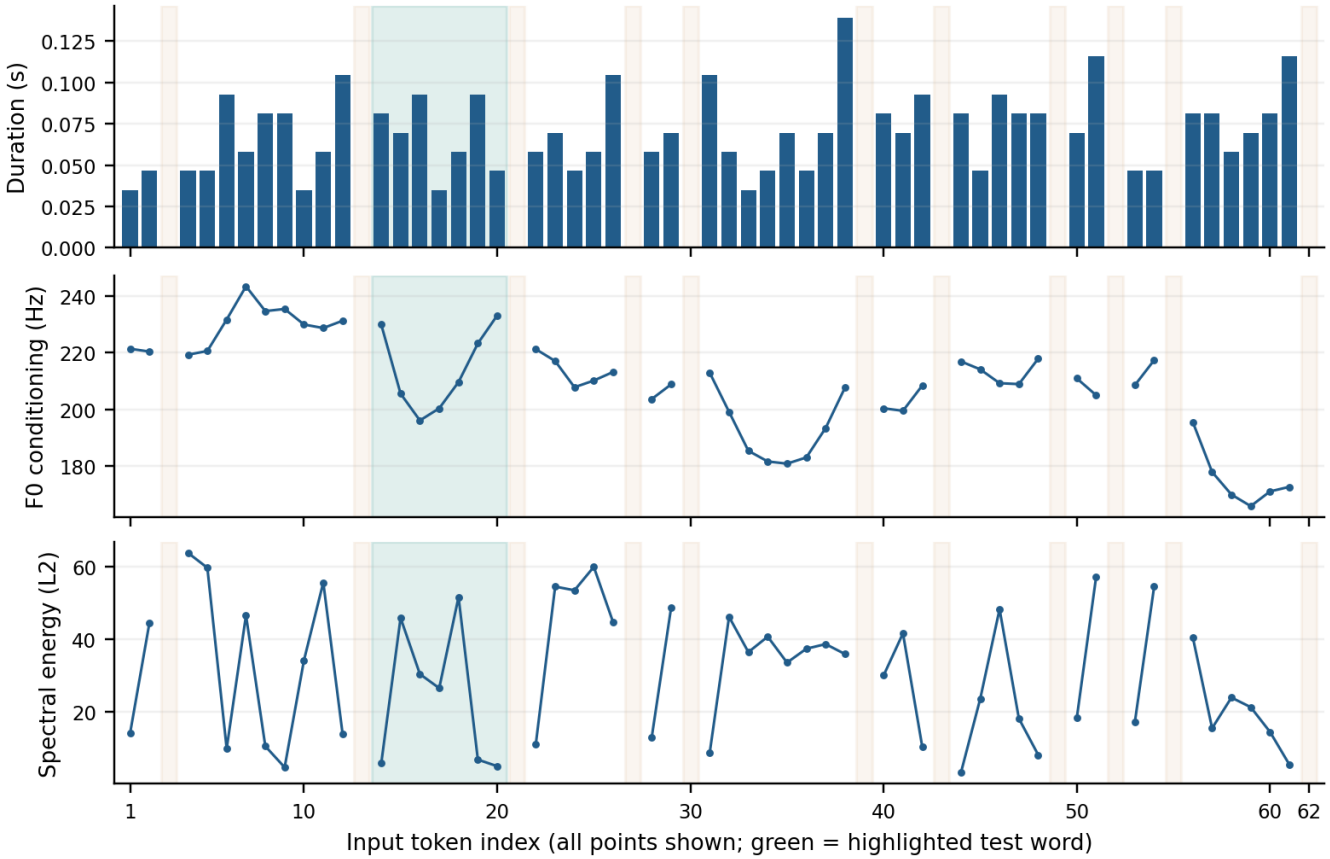
ASR diagnostic: The anti-disastergism-mentarianism movement gained considerable momentum during the parliamentary proceedings.

Files: Results/English/test_generated_audios/E1.wav, .txt, .json, _predictions.npz and _variance.csv.

5.2 E2a | Conduct as a noun

The orchestra's conduct during the ceremony was praised by the audience.

3.587 s | 309 frames | 51 lexical phones | RMS -21.9 dBFS



All model predictions are plotted by token index. Green spans highlight the test word(s); faint orange marks English boundary slots. F0 and energy omit zero-duration tokens. Full phone-labeled originals and CSVs remain in Results/.

The frontend identifies conduct as a noun and supplies K AA1 N D AH0 K T. The seven phones occupy 0.476 s; their F0 conditioning averages 214.2 Hz and spectral energy 23.6.

E2a has no positive internal boundary pause. Its energy curve varies strongly across phones and declines at the end; no inference-time pause was added to make it sound smoother.

The recognizer writes orchestras rather than orchestra's. The 9.1% edit rate is one orthographic possessive mismatch, so it cannot be interpreted as a failed audible consonant or a stress judgment.

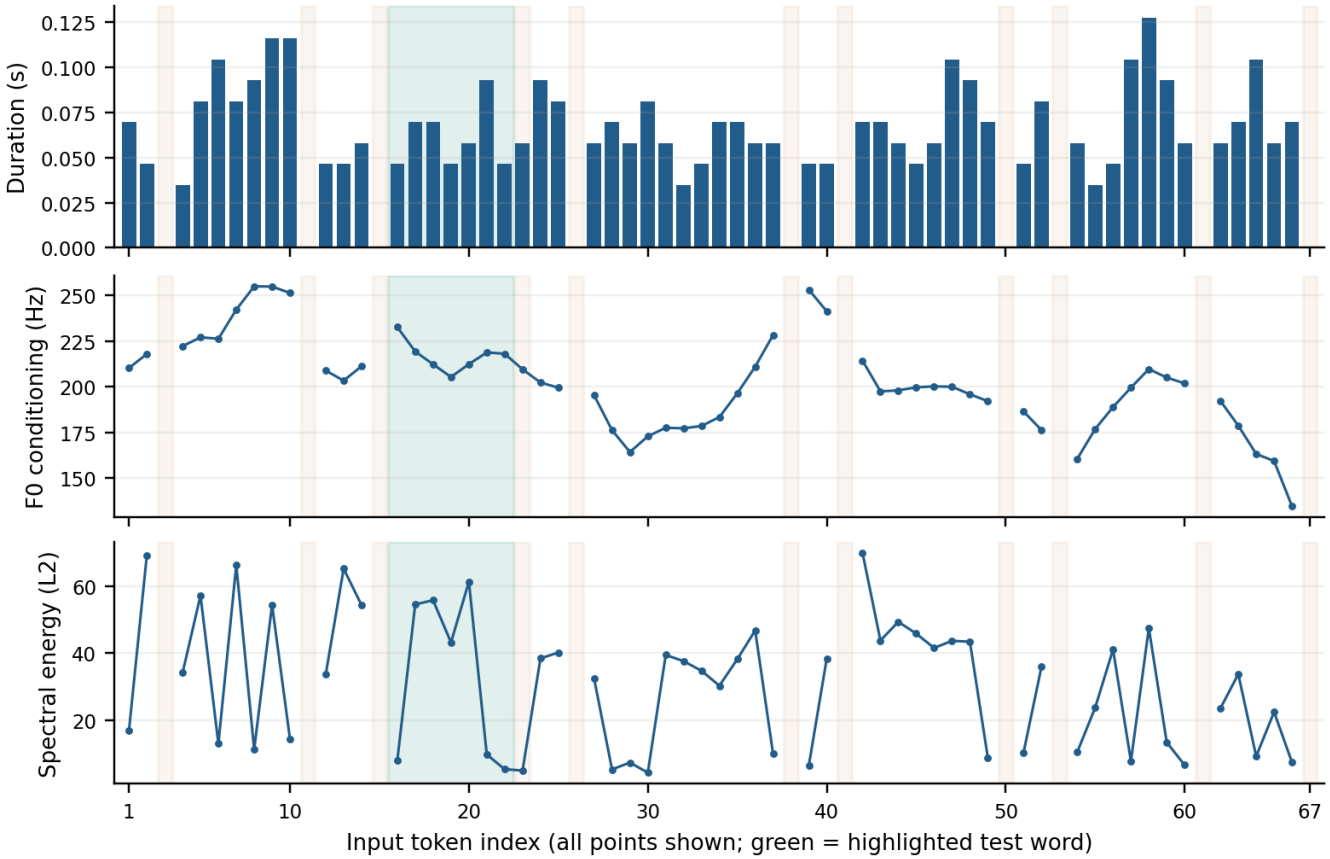
ASR diagnostic: The orchestras conduct during the ceremony was praised by the audience.

Files: Results/English/test_generated_audios/E2a.wav, .txt, .json, _predictions.npz and _variance.csv.

5.3 E2b | Conduct as a verb

The researchers will conduct an experiment to evaluate the proposed method.

3.889 s | 335 frames | 56 lexical phones | RMS -21.3 dBFS



All model predictions are plotted by token index. Green spans highlight the test word(s); faint orange marks English boundary slots. F0 and energy omit zero-duration tokens. Full phone-labeled originals and CSVs remain in Results/.

Following will, conduct is tagged as a verb and encoded K AH0 N D AH1 K T. It occupies 0.430 s versus 0.476 s in E2a. Weighted F0 is 217.0 Hz and spectral energy 34.4; these are contextual outputs under different phone stress and sentence positions.

The learned model predicts one short boundary pause in this sentence. A different total word duration or F0 does not establish semantic understanding; the noun/verb stress contrast was supplied by the text frontend.

Recognition reproduces all normalized words, giving 0/11 edits. This supports lexical intelligibility under this recognizer, while naturalness, stress realization and speaker identity remain unscored.

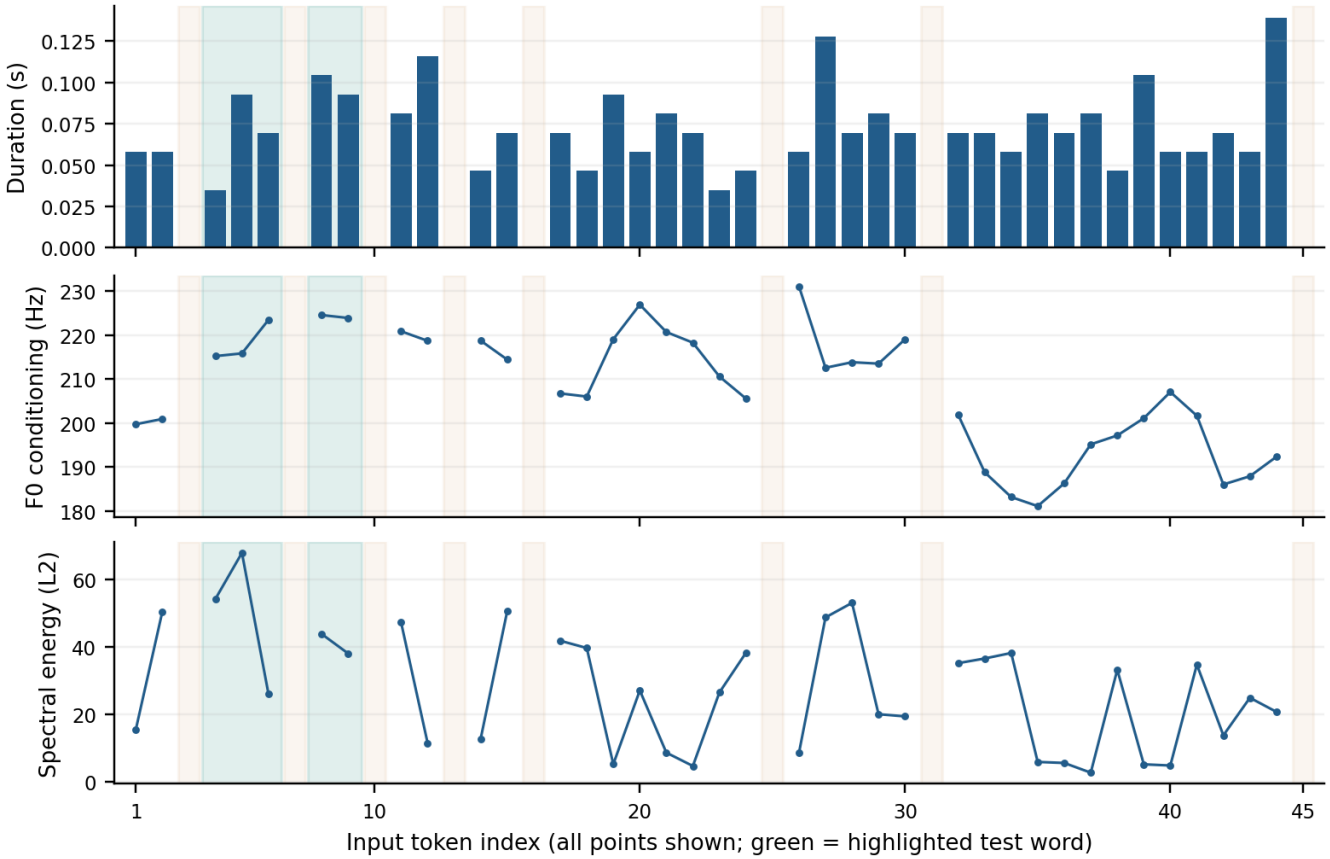
ASR diagnostic: The researchers will conduct an experiment to evaluate the proposed method.

Files: Results/English/test_generated_audios/E2b.wav, .txt, .json, _predictions.npz and _variance.csv.

5.4 E3 | Visible split word

The rhyth m of the orchestra changed unexpectedly.

2.694 s | 232 frames | 37 lexical phones | RMS -22.0 dBFS



All model predictions are plotted by token index. Green spans highlight the test word(s); faint orange marks English boundary slots. F0 and energy omit zero-duration tokens. Full phone-labeled originals and CSVs remain in Results/.

The supplied visible text is rhyth m, not a silently repaired rhythm. The OOV fallback maps rhyth to R AY1 DH; m is separately encoded EH1 M. Each part is predicted for 0.197 s.

No positive boundary pause is produced. The duration-weighted spectral energies of rhyth and m are 50.7 and 41.1, respectively. F0 varies over 181-231 Hz across the utterance; valid phone coverage does not establish the intended conventional pronunciation.

The recognizer returns right end in the corresponding region. Its 25.0% disagreement is sensitive to the unusual source segmentation; it is evidence of a problematic test, not a validated phonetic error count.

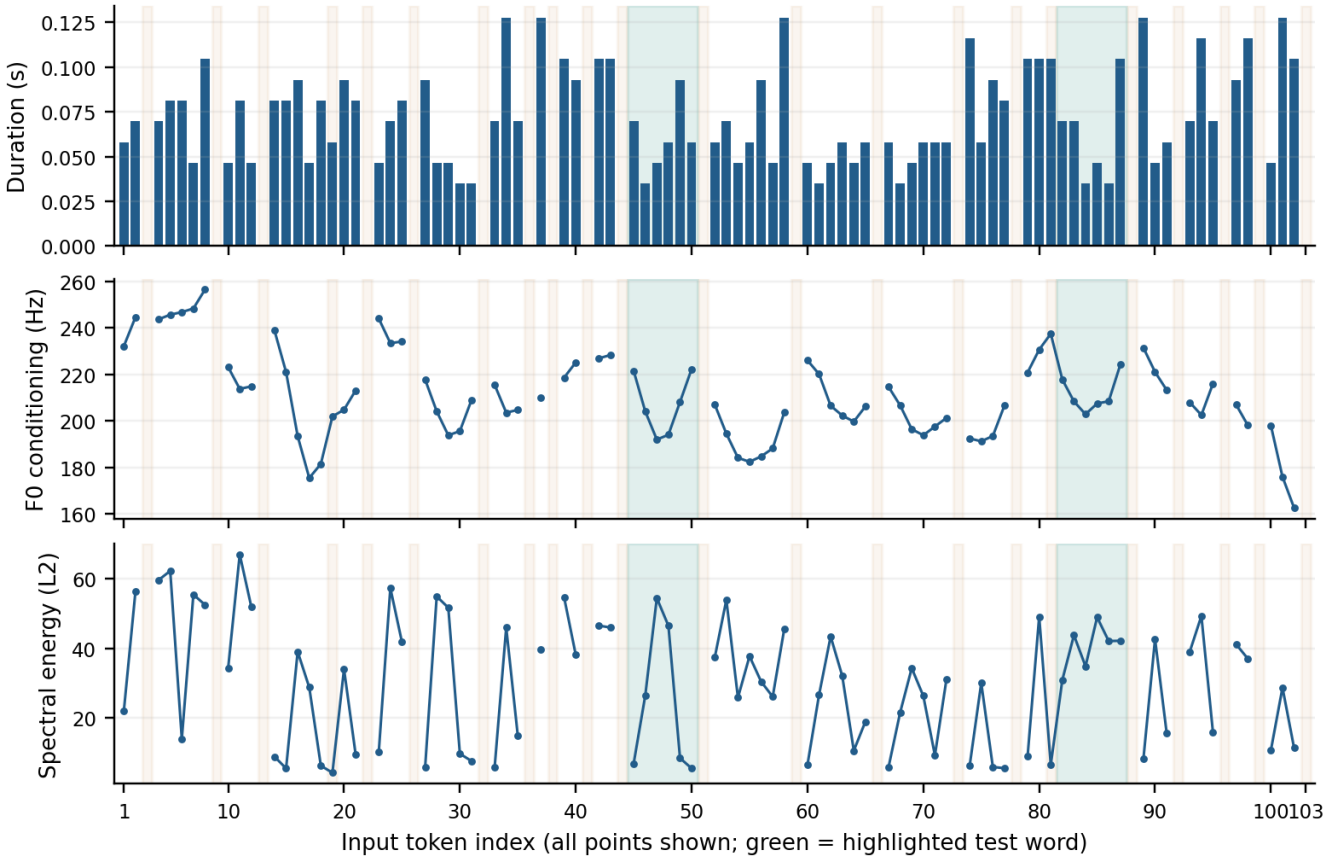
ASR diagnostic: The right end of the orchestra changed unexpectedly.

Files: Results/English/test_generated_audios/E3.wav, .txt, .json, _predictions.npz and _variance.csv.

5.5 E4 | Date, clock and code-mixing

The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

6.014 s | 518 frames | 81 lexical phones | RMS -21.3 dBFS



All model predictions are plotted by token index. Green spans highlight the test word(s); faint orange marks English boundary slots. F0 and energy omit zero-duration tokens. Full phone-labeled originals and CSVs remain in Results/.

Spoken normalization expands 10:45 AM, the ordinal day, and 2026; the Latin andaroo token uses a documented English-phone approximation. The output contains 22 normalized spoken tokens in 6.014 s (219.5 tokens/min), faster than the other English test cases under this tokenization.

The comma after the date receives zero pause frames. Two non-punctuation boundaries contribute 0.163 s total. The andaroo span occupies 0.360 s with weighted F0 conditioning 214.1 Hz and spectral energy 40.5, versus 21.5 for twelfth. Correct intended Telugu articulation is not established.

The recognizer compresses the time/date into a decimal-like number string and changes several nearby words. The 54.5% normalized edit rate flags this as the highest-priority English review case; number-format normalization and ASR failure also contribute uncertainty.

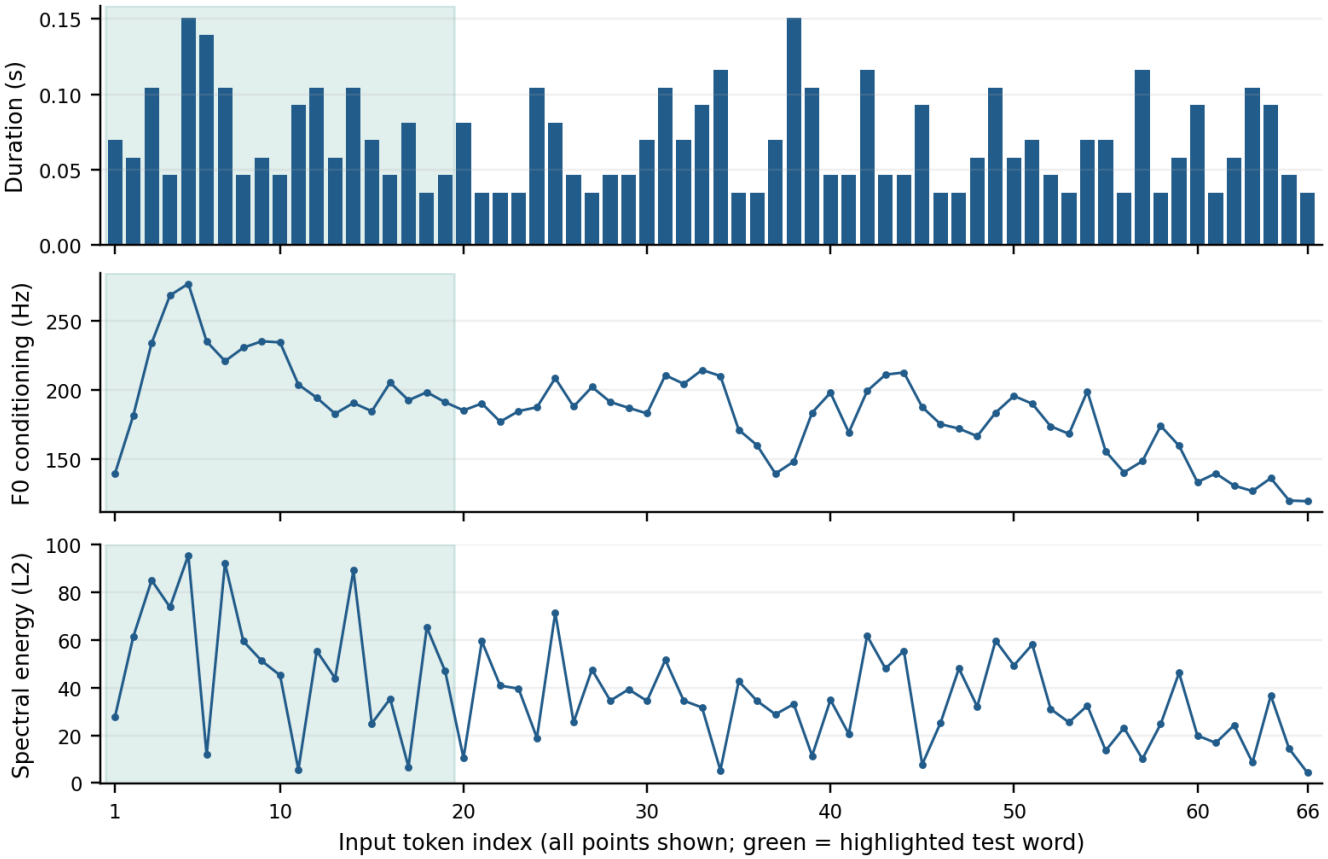
ASR diagnostic: The meeting will start at 10.45.11.12.22.26. Soot and the rule should arrive on that.

Files: Results/English/test_generated_audios/E4.wav, .txt, .json, _predictions.npz and _variance.csv.

5.6 T1 | Long Telugu technical term

న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

4.540 s | 391 frames | 66 lexical phones | RMS -20.0 dBFS



All model predictions are plotted by token index. Green spans highlight the test word(s); faint orange marks English boundary slots. F0 and energy omit zero-duration tokens. Full phone-labeled originals and CSVs remain in Results/.

The technical first word is out of the training lexicon. Telugu eSpeak supplies 19 phones, all mapped into the existing frozen inventory. Its span is 1.463 s of the 4.540 s output; no unknown phone is dropped or newly embedded.

The target region reaches approximately 277 Hz F0 conditioning, then the utterance declines toward 119 Hz. The approximately 74 Hz first-to-last-quintile decline and alternating spectral energy show modeled variation, not a human judgment of expressiveness.

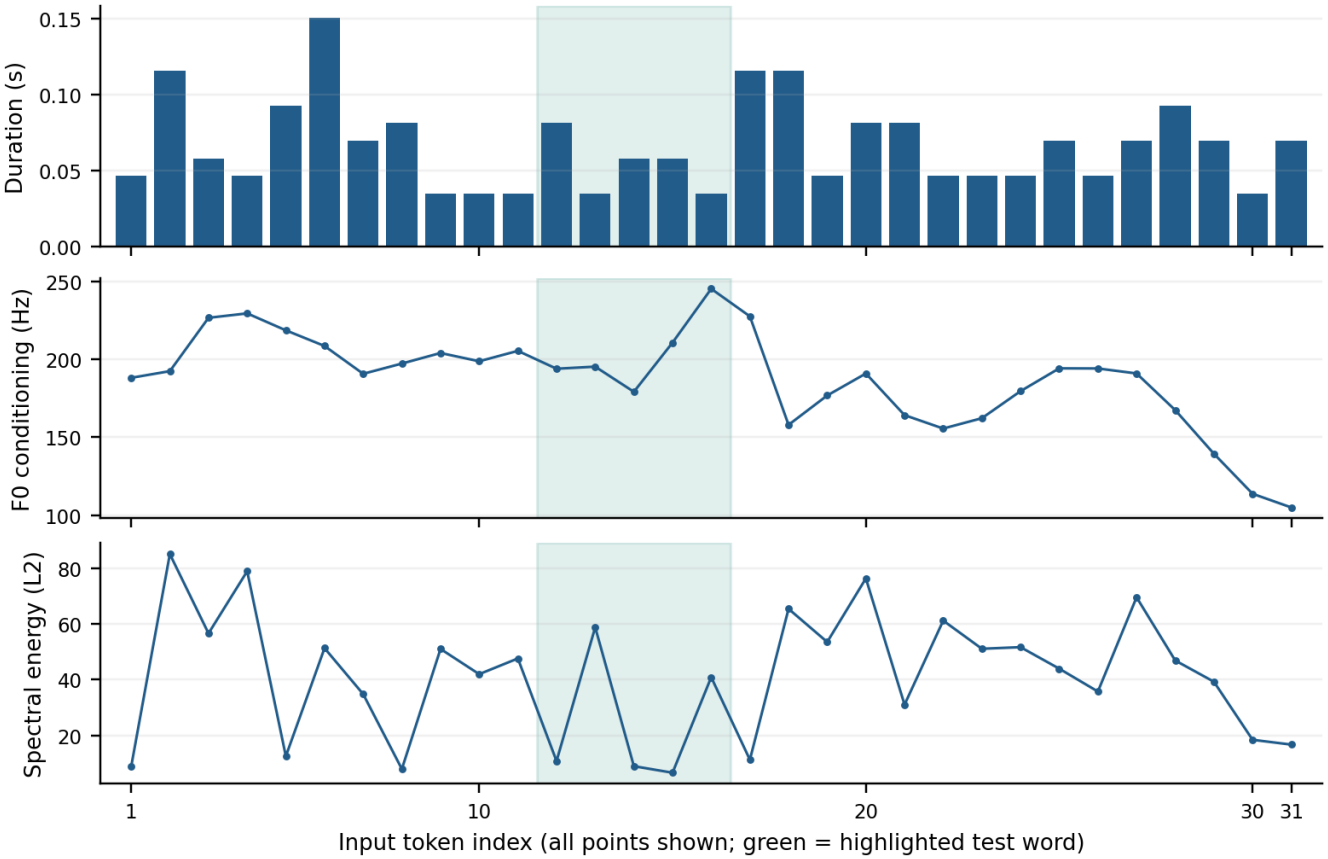
Automatic technical-term pronunciation and aspirated/retroflex distinctions require native listening. No Telugu recognition score is used; signal validity alone cannot establish correct medical terminology.

Files: Results/Telugu/test_generated_audios/T1.wav, .txt, .json, _predictions.npz and _variance.csv.

5.7 T2a | Pattu in the grip context

తాడుపై అతని పట్టు చాలా బలంగా ఉంది.

2.067 s | 178 frames | 31 lexical phones | RMS -19.6 dBFS



All model predictions are plotted by token index. Green spans highlight the test word(s); faint orange marks English boundary slots. F0 and energy omit zero-duration tokens. Full phone-labeled originals and CSVs remain in Results/.

The shared target pattu is encoded /p a retroflex-t retroflex-t u/ using the same five trained phone IDs in both T2 cases. Here its total is 23 frames (0.267 s); weighted F0 conditioning is 201.4 Hz and spectral energy 19.6.

The paired retroflex tokens receive 5+5 frames. The curves show local context effects and a final F0 descent, but the base Telugu frontend supplies no explicit word-boundary pause slots.

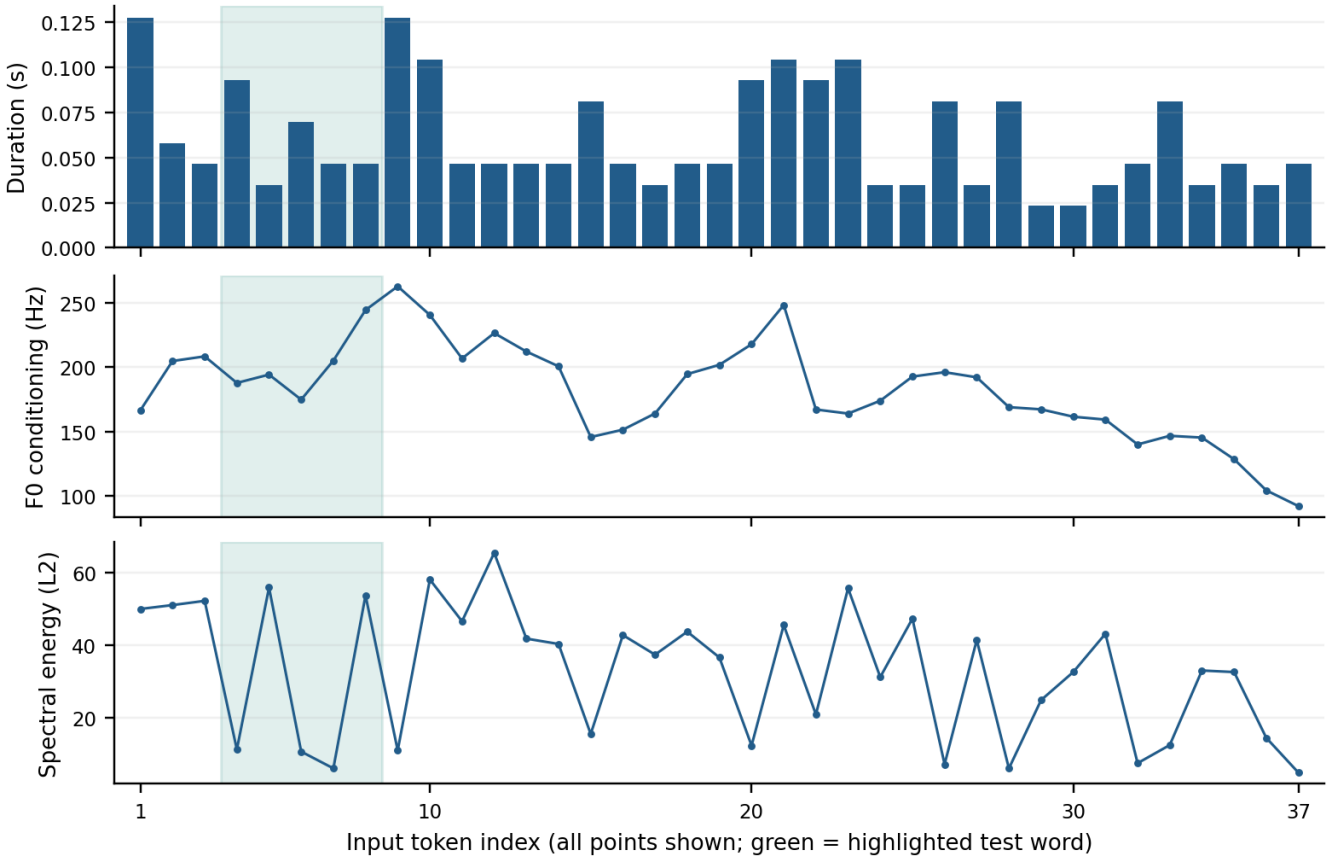
The source concerns grip/hold. Correct lexical meaning, gemination and sentence-level clarity must be checked by a Telugu listener; none is inferred from the phoneme trace alone.

Files: Results/Telugu/test_generated_audios/T2a.wav, .txt, .json, _predictions.npz and _variance.csv.

5.8 T2b | Pattu in the silk context

ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

2.229 s | 192 frames | 37 lexical phones | RMS -21.9 dBFS



All model predictions are plotted by token index. Green spans highlight the test word(s); faint orange marks English boundary slots. F0 and energy omit zero-duration tokens. Full phone-labeled originals and CSVs remain in Results/.

The same pattu phones occupy 25 frames (0.290 s), 23 ms longer than in T2a. Weighted F0 conditioning is 197.1 Hz and spectral energy 22.5. The paired retroflex tokens receive 6+4 frames.

The model changes the word's prosody across the silk-sari and grip contexts, although word position and neighboring phones also change. The small contrast cannot isolate semantic disambiguation or prove correct stress.

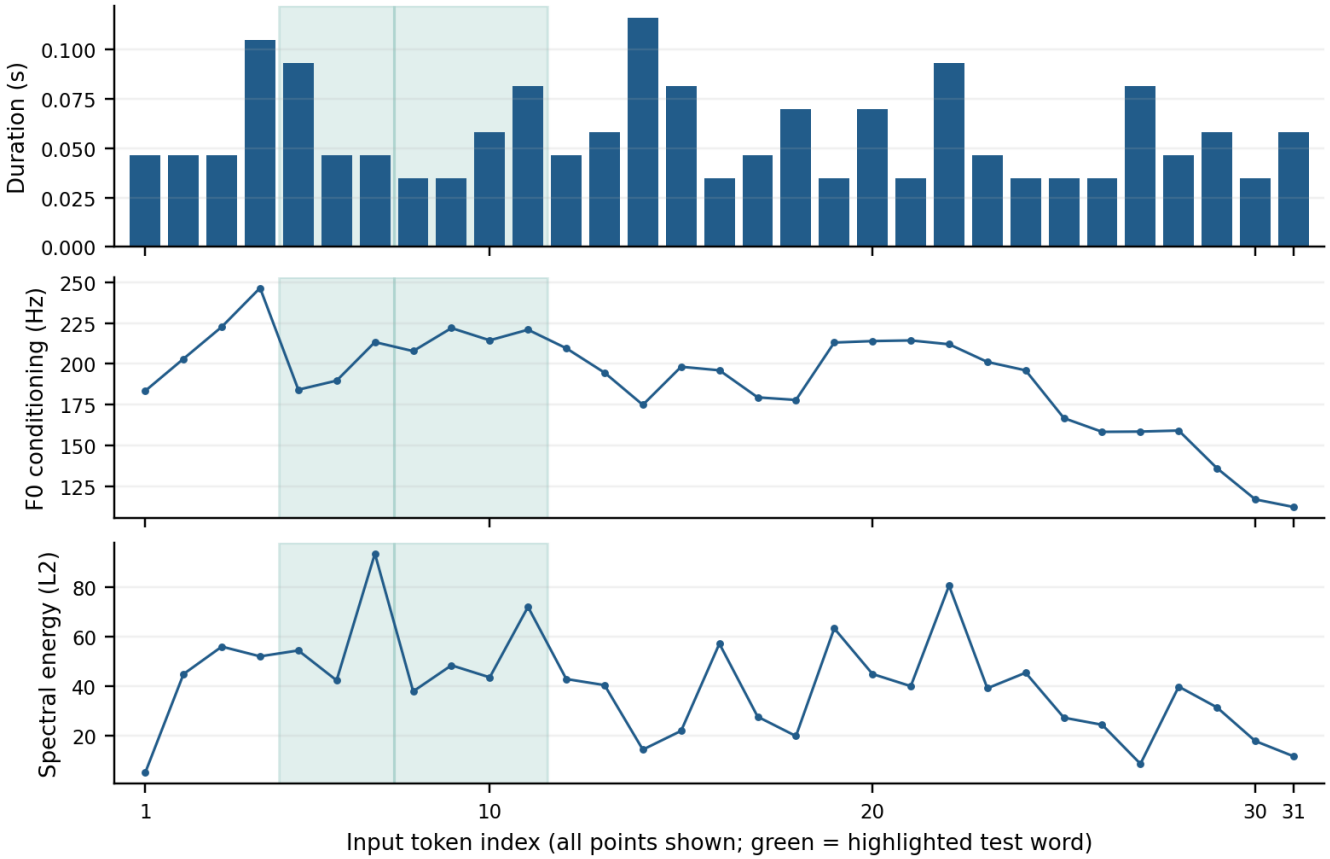
The decoded waveform has no clipping. Native listening remains necessary for the silk-context reading and inflection; the report does not assign an unsupported Telugu pronunciation or MOS score.

Files: Results/Telugu/test_generated_audios/T2b.wav, .txt, .json, _predictions.npz and _variance.csv.

5.9 T3 | Visible split Telugu word

పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

1.753 s | 151 frames | 31 lexical phones | RMS -20.2 dBFS



All model predictions are plotted by token index. Green spans highlight the test word(s); faint orange marks English boundary slots. F0 and energy omit zero-duration tokens. Full phone-labeled originals and CSVs remain in Results/.

The visible source split ada/vilo is preserved as two input words. Their predicted spans are 0.186 s and 0.209 s, with weighted spectral energies 61.1 and 54.4. Both use automatic Telugu pronunciations; the conventional joined spelling is not substituted.

This is the shortest Telugu test at 1.753 s. F0 conditioning falls about 65 Hz from the first to last lexical-phone quintile. Its high-band power fraction is 1.02%, the largest in this set; sibilants can legitimately create that energy, so it is not labeled hiss or noise.

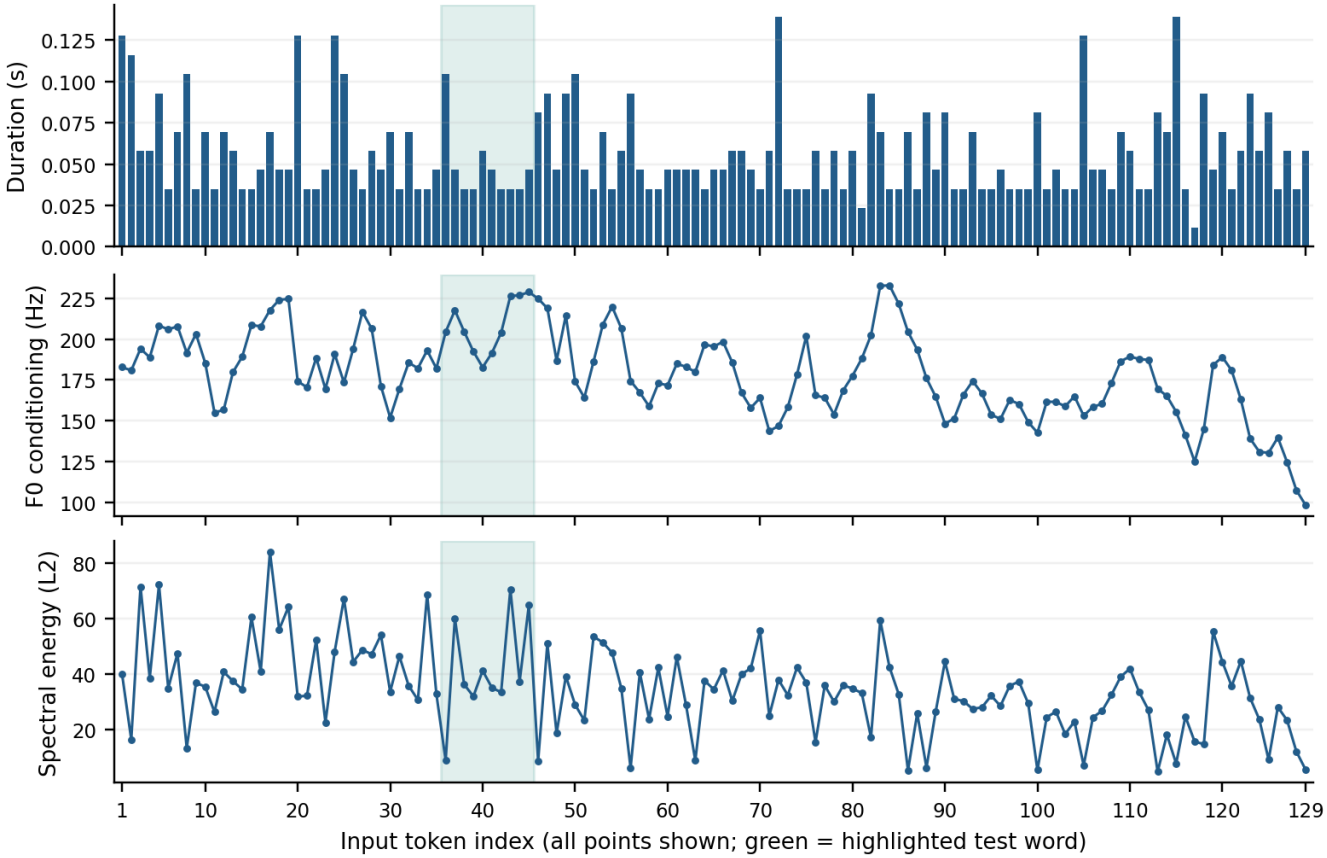
A native listener should distinguish source-spacing effects from acoustic pronunciation and pacing. No recognition-based Telugu quality claim is made.

Files: Results/Telugu/test_generated_audios/T3.wav, .txt, .json, _predictions.npz and _variance.csv.

5.10 T4 | Telugu date and clock

ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

7.233 s | 623 frames | 129 lexical phones | RMS -21.4 dBFS



All model predictions are plotted by token index. Green spans highlight the test word(s); faint orange marks English boundary slots. F0 and energy omit zero-duration tokens. Full phone-labeled originals and CSVs remain in Results/.

The clock phrase is verbalized with hour/minute units, 12va becomes the inflected date ordinal, and 2026 becomes a cardinal year. Source text remains unchanged in the .txt and JSON files. The frozen frontend rejects unsupported numeric ordinals rather than silently inventing them.

All 129 input phones receive frames in the 7.233 s output. F0 conditioning ranges 99-233 Hz, ending lower; energy also falls in the last phones. The base Telugu frontend supplies no explicit comma pause, so punctuation-driven timing is a known limitation.

The complete date/time sequence is represented by trained phones, but correct spoken values and natural phrasing still require native review. No word-accuracy percentage is fabricated for this sentence.

Files: Results/Telugu/test_generated_audios/T4.wav, .txt, .json, _predictions.npz and _variance.csv.

6. Discussion and conclusions

Aspect	English final model	Telugu final model
Data / voice conditioning	Svarah; 254 estimated labels; one cluster used for tests	IndicTTS male subset; one source label
Final architecture	FastSpeech2 plus learned boundaries and pause heads	Base FastSpeech2 with Telugu phone map
Validation result	3.0544 under new seven-term objective	2.1065 under base five-term objective
Observed text-sensitive issue	Long word, split word and mixed date/time recognition	Automatic OOV pronunciation and visible split-word handling need native review
Punctuation timing	Pause detection F1 45.6%; E4 comma 0 frames	No explicit punctuation pause slots in final frontend
Quality evidence	Five ASR diagnostics; no MOS panel	Signals/variance only; recognizer control failed

The two language totals cannot rank language quality. They differ in loss terms, masking, dataset content, speaker composition and normalization. Lower validation loss, a smooth variance curve, non-clipped WAVs and exact ASR text each address different properties; none alone demonstrates natural speech.

Conclusions

The system is operational, source-preserving and reproducible for the ten tests. It predicts nonconstant phoneme-level variance, uses frontend-supplied conduct stress and changes pattu predictions across contexts. Validation plateaus favor earlier checkpoints. English pause F1 is 45.6% versus 38.5% and 21.0% for simple occurrence baselines; selected pause-duration MAE is slightly worse than initialization. These results establish limited predictor-level gains, while audible prosody improvement remains unproven.

Limitations and future work

First, collect native per-test listening observations using the existing review page and preserve those observations separately from automated scores. Audit OOV and number/cross-language pronunciations, especially E1/E3/E4 and T1/T3/T4. Improve speaker-label confidence and coherent voice coverage before assuming more English steps will fix clarity. Rebalance informative punctuation examples and validate occurrence calibration and pause timing independently; held-out clause/exclamation coverage is currently insufficient.

Only after those checks should another bounded pilot be considered. T4 has identical phonemes and all-zero boundary inputs with or without its comma: Telugu punctuation cannot condition the current acoustic output. Fixing this requires consistent training and inference supervision. Finer pitch trajectories, a voiced/unvoiced model or a better-matched vocoder are also unmeasured future experiments. Appendix D quantifies speaker sparsity and the remaining quality-evaluation limits.

Appendix A. Reproducibility

Run commands from the repository root. Use an isolated Python 3.10 environment. src/setup/requirements-inference.txt pins inference packages; root requirements.txt combines preprocessing, training and inference dependencies. Recorded GPU training used PyTorch 2.8.0+cu128. The short README covers inference; src/docs/REPRODUCIBILITY.md gives CUDA wheels, MFA setup, preprocessing, training and verification commands.

```
uv venv --python 3.10 .venvs/final-inference
uv pip install --python .venvs/final-inference/bin/python -r src/setup/requirements-inference.txt
python3 src/setup/download_checkpoints.py
.venvs/final-inference/bin/python src/setup/prepare_final_inference.py --nltk-only
.venvs/final-inference/bin/python English/inference.py --test-id E1
.venvs/final-inference/bin/python Telugu/inference.py --test-id T1
.venvs/final-inference/bin/python src/setup/generate_assignment_outputs.py
.venvs/final-inference/bin/python src/setup/verify_assignment_outputs.py
```

Preprocessing in a clean reconstruction

The reproduction guide lists pinned dataset-audit commands, src/setup/restore_assignment_splits.py and export_split_audio.py to restore source membership and frozen speaker labels after authorized dataset access. Linux alignment expects MFA 3.3.9 at /workspace/.venvs/mfa and english_us_arpa.zip under /workspace/.cache/mfa/pretrained_models/acoustic/. These wrappers build corpora, alignments and corrected features, then check a real batch:

```
python English/preprocess.py --stage all --device cpu --feature-version gap-fixed-v2
python Telugu/preprocess.py --stage all --device cpu --feature-version gap-fixed-v2
python src/setup/build_learned_pause_features.py
```

The last command builds measured pause features and the three training YAMLs. For existing features, add --configs-only to reconstruct the YAMLs without rewriting targets. The committed src/config/learned-pauses-v1/English/ snapshot records the original experiment. Dataset access follows source license/gate conditions.

Training reconstruction

```
python English/train.py --run-id english-new --fresh --minutes 120 --max-steps 30000 --batch-size 8
python Telugu/train.py --run-id telugu-new --fresh --minutes 120 --max-steps 30000 --batch-size 8
```

These commands start new CUDA runs. Fine-tuning adds --initialize-from, its matching --initialize-sha256, --learning-rate 2e-5 and patience 5. Pause training adds --learned-pauses, its generated --config-root, --new-head-learning-rate 2e-4 and patience 8. Resume with --resume and the run ID, without initialization flags. A retained session/STOP is rejected before writes; remove only that run's marker after explicitly deciding to continue. Policy, optimizer and best selection are restored. Explicit --device cpu supports local tests.

Appendix B. Checkpoints and artifacts

The final English weights are learned-pause step +1,000 on fine-tune step +2,500, itself initialized from corrected step 12,000. Telugu uses corrected step 18,000. Parent checkpoints remain separate. SHA256 fingerprints identify the exact selected local files:

English/checkpoints/final.pth.tar

`74ce8b28183ec37f98a001bfcd5b484cb660a2eea2d1a780ef6dc8ea7e6a38bc`

Telugu/checkpoints/final.pth.tar

`2268f94c05c0c90d9a4b85780971b40dd07297bc39a16271b375311d4b7fcfb8`

Artifact	Location
Frozen weights / metadata	English/checkpoints/ and Telugu/checkpoints/
Inference configs/resources	English/config.yaml, Telugu/config.yaml; resources/ in each
Audios, source text, predictions	Results/{English,Telugu}/test_generated_audios/
Thirty full-label variance figures	Results/{English,Telugu}/variance_adaptor_plots/
Final selected-run loss plots	Results/{English,Telugu}/{training,validation}_loss.png
Interactive local review / output ZIP	Results/Assignment/index.html; assignment_test_outputs.zip
Measured analyses / recognition	Results/Assignment/analysis/
Report source and figure scripts	src/setup/build_technical_report.py; create_report_figures.py

The repository root is the assignment's project/ directory: English/, Telugu/, Results/, Report/, README.md and requirements.txt. Shared modules and tools are grouped under src/; original upstream demos are isolated in src/reference/. Historical training records retain their original paths and hashes. Current commands above use the reorganized source paths.

The assignment release contains a complete project/ archive with separate English/Telugu model-only checkpoints and the frozen vocoder. SHA256SUMS verifies the archive; src/setup/download_checkpoints.py verifies each restored model. Per-language training wrappers, updated requirements and the README are included. The repository URL remains to be submitted through the assignment form.

Corrected assignment-v5 release: complete project and checkpoint download

GitHub repository: ParichayMago/fastspeech2-english-telugu

Recreate the report without cloud compute

```
.venvs/inference/bin/python src/setup/analyze_assignment_outputs.py
```

```
.venvs/inference/bin/python src/setup/create_report_figures.py
```

```
.venvs/report/bin/python src/setup/build_technical_report.py
```

The report builder consumes the saved English recognition cache. Optional analysis flag --asr computes missing or changed Whisper-small results and requires openai-whisper==20250625. Report dependencies are pinned in Report/requirements-report.txt. The included Noto Sans Telugu font is shaped with HarfBuzz and distributed under the accompanying SIL Open Font License.

Appendix C. Post-review corrections

Release assignment-v5 includes the prior recovery/portability fixes and corrects retained-STOP handling and final-recording discovery. Historical logs, weights, ten test WAVs and predicted arrays are preserved. These software fixes require no retraining and do not establish clearer speech.

Correction	Verified behavior
Best-checkpoint preservation	Restore historical best loss/hash and stale-validation count; worse evaluations cannot replace best; missing/corrupt best files fail closed.
Base and fine-tune recovery	Restore the saved model, configuration and Adam state, including fixed learning-rate groups and early stopping. Resume does not reread the original parent checkpoint.
Pause configurations	Feature construction writes all three YAMLs from measured training statistics. --configs-only reconstructs them; original configs are also distributed.
Tensor device allocation	Masks use lengths.device and length-regulator output uses x.device, independent of CUDA availability.
Energy control	Both phoneme-level and frame-level energy paths receive e_control. Unity-control generation remains identical.
Review interface	Discover all ten final WAV/JSON pairs, original transcripts and current hashes; export all reviewed statuses. Saved judgments remain local.
Retained stop request	Reject explicit/implicit resume before writes when session/STOP remains; never clear it automatically. New in-run stop requests still save gracefully.

Local validation and its scope

The local setup suite passes 36 tests and the review server passes nine: 45 total. Synthetic CPU fixtures exercise base, fixed-rate and pause recovery, legacy checkpoints, Adam continuation, historical best/config restoration, retained-STOP rejection without writes and graceful in-run stopping. Real-data pause integration also passes locally. Browser checks cover audio playback, save/reload/export, draft retention and narrow-screen layout; synthetic ratings use temporary stores, not listener data.

All ten frozen sentences regenerate with identical WAV SHA256 values and exact equality of every exported array, including mels and pause-head outputs. No released model parameters were updated. No Runpod resource was started or modified. CUDA-host execution and a fresh dependency installation were not repeated; CPU device tests simulate CUDA availability and include meta-device allocation.

Reproduce the correction checks

```
python -m unittest discover -s src/setup -p test_review_fixes.py -v
```

```
python -m unittest discover -s src/audio_review -p test_server.py -v
```

```
python src/setup/verify_frozen_regeneration.py
```

```
python src/setup/evaluate_pause_baselines.py
```

Recovery tests require the full requirements.txt stack, including TensorBoard. The baseline command needs local held-out pause targets; its saved result is distributed. Real-data integration explicitly skips when private fixtures are absent; skips are not passes. Per-sentence listening observations, especially Telugu pronunciation, remain outstanding. MOS scores and audible pause improvements are not asserted.

Appendix D. Speaker coverage and remaining limits

Estimated English label coverage

Measure	Selected source	Accepted source
Training clips / estimated labels	3,039 / 254	3,037 / 254
Labels with 1-2 clips	77 (30.3%)	77 (30.3%)
Median clips / source seconds per label	7 / 38.3 s	7 / 38.3 s
Final label en_est_0155: clips	90	89
Final label: source duration	7.500 min	7.482 min

These durations sum source WAV lengths, including edges, not cropped acoustic features or non-silent speech. Two selected clips fail feature preparation. Shared English parameters learn from the full approximately five-hour selection across all labels; 7.5 minutes describes one conditioning label, not the entire English corpus. Accepted aligned feature hours remain 4.6536 as reported in Section 2.1.

ECAPA clustering uses training recordings with at least 3 s of speech, average-linkage cosine distance and an uncalibrated 0.30 threshold. Short training and validation recordings use frozen training centroids. The publisher reports 117 speakers [4], while this pipeline estimates 254 labels. That mismatch warrants a fragmentation/merge audit; it cannot establish exact identity errors or causation for poor English speech.

The final label is third by selected source duration and is retained for continuity with earlier previews. No systematic perceptual ranking of labels was recorded. Sparse per-label coverage and uncertain membership are plausible conditioning risks, not experimentally isolated causes. A verified label map or coherent voice subset would change the modeling experiment and require new training/evaluation.

Remaining evaluation and modeling limits

Pronunciation, clarity and overall-quality observations for every frozen test recording remain incomplete, particularly for Telugu. Ratings on earlier previews do not substitute for those observations. ASR disagreement (including E4 at 54.5%) flags investigation but does not identify exact audible errors. No MOS or audible parent-versus-pause improvement is established.

English pause occurrence improves against simple baselines, but 1,705 false positives exceed 1,022 true positives; duration MAE worsens and the E4 comma has zero frames. Telugu removes punctuation and supplies no pause slots despite internal silence in training; T4 inputs are unchanged when its comma is removed. The review-server and STOP fixes address software behavior, not these acoustic-model limitations.

```
python src/setup/analyze_speaker_coverage.py
```

Recomputes coverage from distributed selection-membership.jsonl, rejected feature IDs and saved clustering metadata. Full per-label counts, durations and input SHA256 values: Results/Assignment/analysis/english-speaker-coverage.json. No raw audio, new training or cloud compute is required.

References

[1] Ren et al. FastSpeech 2: Fast and High-Quality End-to-End Text to Speech. ICLR 2021.

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[2] ming024. FastSpeech2 PyTorch implementation; base checkout d4e79eb52e8b01d24703b2dfc0385544092958f3.

https://github.com/ming024/FastSpeech2

[3] Kong, Kim and Bae. HiFi-GAN: Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis. NeurIPS 2020.

https://arxiv.org/abs/2010.05646

[4] AI4Bharat. Svarah dataset card and access conditions.

https://huggingface.co/datasets/ai4bharat/Svarah

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[6] Montreal Forced Aligner 3.3.9 documentation.

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[7] Kyubyong/g2p. g2p-en text, homograph and pronunciation implementation.

https://github.com/Kyubyong/g2p

[8] Phonemizer and eSpeak-NG loader documentation.

https://pypi.org/project/espeakng-loader/

[9] SpeechBrain. Pretrained ECAPA-TDNN speaker embedding model.

https://huggingface.co/speechbrain/spkrec-ecapa-voxceleb

[10] Radford et al. Robust Speech Recognition via Large-Scale Weak Supervision.

https://arxiv.org/abs/2212.04356

[11] Seed-TTS (2024). A Family of High-Quality Versatile Speech Generation Models. Technical report; reference for report organization.

https://arxiv.org/abs/2406.02430

Project records, 4 November 2026

Dataset revisions, split integrity and hours: Results/DatasetAudit/; Results/DataSplits/summary.json and verification.json.

Corrected feature integrity and restored gaps: Results/Preprocessing/gap-fixed-v2/rebuild-summary.json.

Optimizer logs: Results/Training/{gap-fixed-v2-20261103,english-finetune-pilot-20261104,english-learned-pauses-pilot-20261104}/<language>/metrics.jsonl; copied unchanged from local output/runs/ backups.

Exact source strings: src/setup/assignment_test_sentences.json; source PDF SHA256 a88e379253c50a018ff6319c05a372a57a0d416cf7c4dc20807d7752e67e323f.

Final output hashes and software checks: Results/Assignment/verification.json; checkpoint-selection.json; CLI smoke verification.

Computed analysis and ASR hypotheses: Results/Assignment/analysis/measured-analysis.json and english-asr.json. All numerical tables and report plots derive from these saved records.

Telugu-recognizer exclusion evidence: Results/InitialSpeechReview/review.json, including natural-Telugu control. No native-listener or panel MOS ratings were available at generation time.